National University of Computer & Emerging Sciences, Karachi
School of Computing
Final Exam, Fall 2022
Monday, 2^{nd} Jan.2023, 12:00 pm– 3:00 pm

Course Code: MT-1004	Course Name: Linear Algebra
Instructor(s): Mr.JamilUsmani, Dr.KhusroMian	
Student Roll No:	Section:

Instructions:

- Return the question paper along with answer sheet.
- Use of Scientific calculator is allowed.
- Read each question completely before answering it. There are 6 questions and 2 pages.
- In case of any ambiguity, you may make assumption. But your assumption should not contradict any statement in the question paper.
- Solve all the questions in sequence and show all necessary steps for full credit.

Time: 180 minutes. **Max Marks**: 50 points

Question 1 [CLO-2] **Marks (2+1+2=05)**

Let $A = \begin{bmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{bmatrix}$

I. Convert matrix A into reduced row echelon form.
II. Write the parametric solution if consistant and find the Rank and nullity of A.
III. Find a basis for the row ,column and null space of A.

Question 2 [CLO-3] **Marks (2+2+6=10)**

Consider $Q = 5x_1^2 + 2x_2^2 + 4x_3^2 + 4x_1x_2$

I. Express the quadratic form in matrix notation X^tAX , where A is symmetric .
II. Find the eigen value of symmetric matrix and identify the conic.
III. Find an orthognal change of variable that eliminate the cross product term in quadratic form Q and express Q in term of new variables.

Question 3 [CLO-3] **Marks (5+3+2=10)**

Consider $A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$ and eigenvectors $u_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, u_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, u_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

I. Use Gram-schmidt process to construct an orthogonal matrix ***P*** from eigen vectors.
II. Find the diagonal matrix D such that $PAP^{\mathrm{T}} = D$
III. Display the matrix A into spectral decomposition that has orthogonaly diagonalize.

Question 4 [CLO-3] **Marks (3+2=05)**

a) Let $U = \begin{bmatrix} 3 & -2 \\ 4 & 8 \end{bmatrix}$, $V = \begin{bmatrix} -1 & 3 \\ 1 & 1 \end{bmatrix}$, Compute $||U||$& $d(U, V)$

and check for orthogonality of matrix relative to the standard inner product on M_{22}

b) Find cosine angle between the vectors $u = (-3,2)$, $v = (1,7)$ with respect to the weighted Eucledian inner product $< u, v >= 2u_1v_1 + 3u_2v_2$ on $\mathcal{R}^2$

Question 5 [CLO-3] **Marks (3+7=10)**

a) Express the matrix *A*= *QR* decomposition ,where *Q* matrix obtained by applying the Gram-schmidt orthonormalization process to the column vector of A

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix}, Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{6}} \end{bmatrix}$$

b) Find the unitary matrix *P* that diagonalize the Hermitian matrix $A = \begin{bmatrix} 3 & -i \\ i & 3 \end{bmatrix}$

and determine $P^{-1}AP$

Question 6 [CLO-1] **Marks (2+2+6=10)**

a) Determine whether the following vectors spans $\mathcal{R}^3$ or not ?

$$u_1 = (2, -1, 3),\ \ u_2 = (4,1, 2)\ \ and\ \ u_3 = (8, -1, 8)$$

b) Let $A = \begin{bmatrix} 1 & -2 & 8 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ and Let $P = \begin{bmatrix} 1 & -4 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

Confirm that ***P*** diagonalize ***A*** and then compute $\boldsymbol{A^{2301}}$

c) Find the LU decomposition of coefficient matrix and solve the system.

$$\begin{bmatrix} -3 & 12 & -6 \\ 1 & -2 & 2 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -33 \\ 7 \\ -1 \end{bmatrix}$$

ALL THE BEST